

SUMMARY

The Boston Public Library has spent **15 years digitizing** historical photographs, newspapers, maps, and manuscripts via **Digital Commonwealth**. Today, users can only search by rigid keyword tags.

The Problem

If an item was never explicitly tagged with the right keyword, it can't be found — even if the answer is right there in the document text.

Our Goal

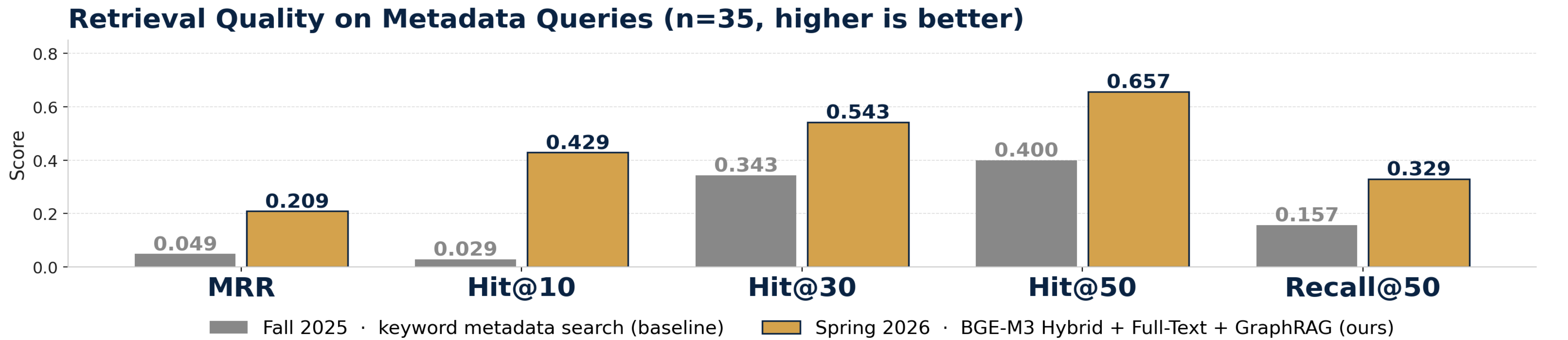
Build a natural-language search system that lets researchers, students, and the public ask plain-English questions and get back relevant historical documents with cited, source-grounded answers.

What We Built

An end-to-end Hybrid + GraphRAG pipeline with cited GPT-4o answers, deployed as a multi-page Streamlit app on Hugging Face — see Page 2 for system architecture.

RESULTS

Retrieval performance vs. Fall 2025 keyword baseline · n = 35 metadata queries from a held-out gold-standard set



Metadata Hit@10

0.029 → **0.429**

≈ 15× improvement

Metadata Hit@50

0.400 → **0.657**

+64% lift

Metadata MRR

0.049 → **0.209**

≈ 4× improvement

CLIENT AND MENTORS

Client
Eben English
Boston Public Library · Digital Commonwealth

Project Managers
Ryan Rodriguez (PM) · Shinu Shibu (TPM)

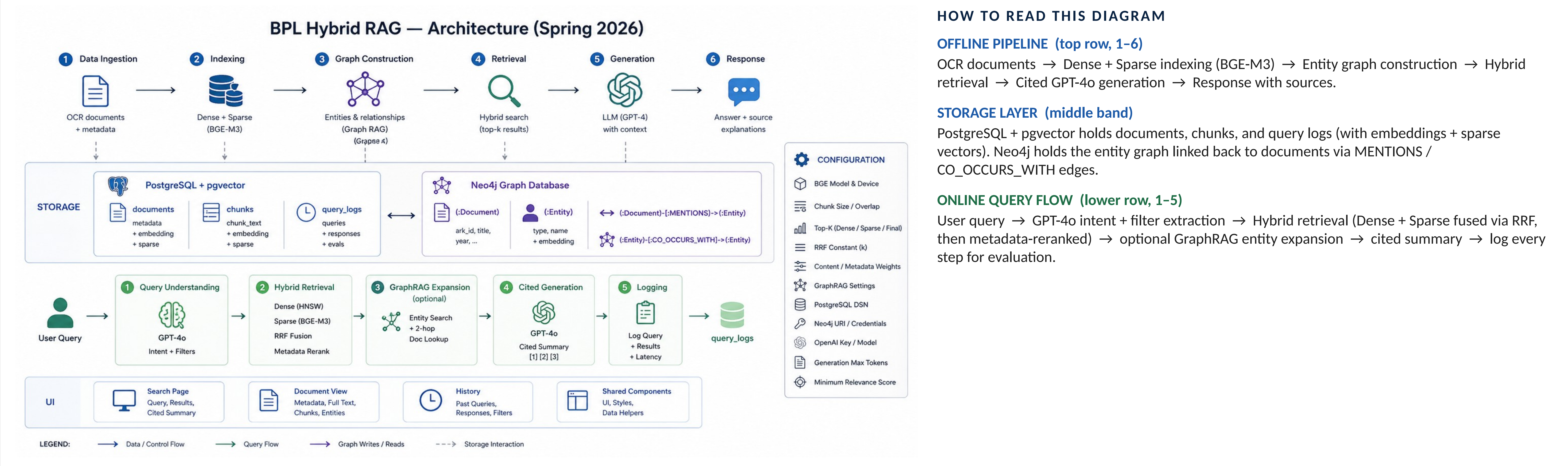
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CONCLUSION & NEXT STEPS

- KEY INSIGHT** Hybrid + Full-Text + GraphRAG retrieves the gold-standard document in **57% of all queries** (Hit@50, n=51) — vs. 39% for the Fall 2025 metadata baseline.
- First full-text search integration in project history — OCR newspaper content now indexed and searchable.
 - End-to-end Hybrid + GraphRAG pipeline with cited GPT-4o answers and structured query logs.
 - Multi-page Streamlit app deployed on Hugging Face — search, document detail, and query history.
 - Next: scale corpus from current subset to the full 1.5M-page collection, add a cross-encoder reranker, and extend to images via CLIP / SigLIP.**

SYSTEM ARCHITECTURE · HOW BPL-LIBRAG WORKS



STORAGE

PostgreSQL + pgvector

documents metadata + embedding + sparse

chunks chunk_text + embedding + sparse

query_logs queries + responses + evals

Neo4j Graph Database

(:Document)

ark_id, title, year, ...

(:Entity)

type, name + embedding

(:Document)-[:MENTIONS]->(:Entity)

(:Entity)-[:CO_OCCURS_WITH]->(:Entity)

1 Query Understanding

GPT-4o Intent + Filters

2 Hybrid Retrieval

Dense (HNSW)
Sparse (BGE-M3)
RRF Fusion
Metadata Rerank

3 GraphRAG Expansion (optional)

Entity Search + 2-hop
Doc Lookup

4 Cited Generation

GPT-4o Cited Summary [1] [2] [3]

5 Logging

Log Query + Results + Latency

query_logs

UI

Search Page
Query, Results, Cited Summary

Document View
Metadata, Full Text, Chunks, Entities

History
Past Queries, Responses, Filters

Shared Components
UI, Styles, Data Helpers

LEGEND:

→ Data / Control Flow

→ Query Flow

→ Graph Writes / Reads

→ Storage Interaction

HOW TO READ THIS DIAGRAM

OFFLINE PIPELINE (top row, 1-6)
OCR documents → Dense + Sparse indexing (BGE-M3) → Entity graph construction → Hybrid retrieval → Cited GPT-4o generation → Response with sources.

STORAGE LAYER (middle band)
PostgreSQL + pgvector holds documents, chunks, and query logs (with embeddings + sparse vectors). Neo4j holds the entity graph linked back to documents via MENTIONS / CO_OCCURS_WITH edges.

ONLINE QUERY FLOW (lower row, 1-5)
User query → GPT-4o intent + filter extraction → Hybrid retrieval (Dense + Sparse fused via RRF, then metadata-reranked) → optional GraphRAG entity expansion → cited summary → log every step for evaluation.

METHODS / TECH STACK

Embeddings

BGE-M3 dense + sparse (1024-d, single forward pass)

Vector Store

PostgreSQL + pgvector (HNSW ANN over dense + sparse)

Hybrid Retrieval

Dense ANN + Sparse lexical, fused via Reciprocal Rank Fusion + metadata rerank

Graph RAG

Neo4j · spaCy NER · 2-hop CO_OCCURS_WITH traversal for entity expansion

Query & Generation

GPT-4o intent classifier + cited summary writer with [1][2][3] inline source markers

Interface

Multi-page Streamlit app · deployed on Hugging Face Spaces

Evaluation

51 gold-standard queries · MRR / Hit@K / Recall@K / Precision@K (K = 10, 30, 50)

REFLECTIONS — WHAT WE LEARNED

A RAG pipeline is a chain

— every model swap meant rebuilding indexes, retuning fusion weights, and re-running the full evaluation from scratch. Plan for re-runs, not one-shots.

Historical OCR is hard for embedding models

— sparse output, broken tokens, and 19th-century vocabulary all hurt dense recall. Sparse retrieval is what saved us on noisy text.

Hybrid > either alone

— dense + sparse fusion materially outperformed either component, especially on metadata-heavy queries: Hit@50 jumped 0.400 → 0.657.

Evaluation comes first, not last

— without a real test set and a harness, every change is a guess. The harness is now the single source of truth for the team.

Citations build trust

— librarians and researchers won't accept ungrounded LLM output. Inline [1][2][3] markers + source links were table-stakes for adoption.